DO NOW — RECALL

Clock speed, cache and pipelining

Name: _____ Class: _____ Date: _____

1. What does the data bus transport during an instruction fetch?

2. Which register holds the address of the next instruction?

3. What is a dependency between tasks?

4. A rising-edge D flip-flop initially has Q=0. D changes 0→1 between edges, then 1→0 just before the next rising edge. What is Q immediately after that edge? Justify.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The instruction's contents from memory towards the CPU.

2. Model answer

PC.

3. Model answer

One operation cannot proceed until another result is ready.

4. Model answer

Q=0. The flip-flop samples D at the rising edge (D=0 at that instant); an earlier pulse between clock edges is not stored.

Vocabulary — know these terms

clock speed — Number of processor clock cycles each second.

cache — Small, fast memory that reduces some main-memory accesses.

pipeline — Overlap stages of several instructions.

hazard — Dependency or uncertainty disrupting smooth pipelining.

throughput — Amount of work completed in a given time.

NEW LEARNING

Read, annotate and explain

Different performance factors

Clock speed indicates cycles per second; multiple cores can execute independent work; cache reduces some slower-memory access.

Overlap pipeline stages

Without overlap, three three-stage instructions need nine stage slots; an ideal pipeline finishes them by the fifth cycle.

Account for hazards

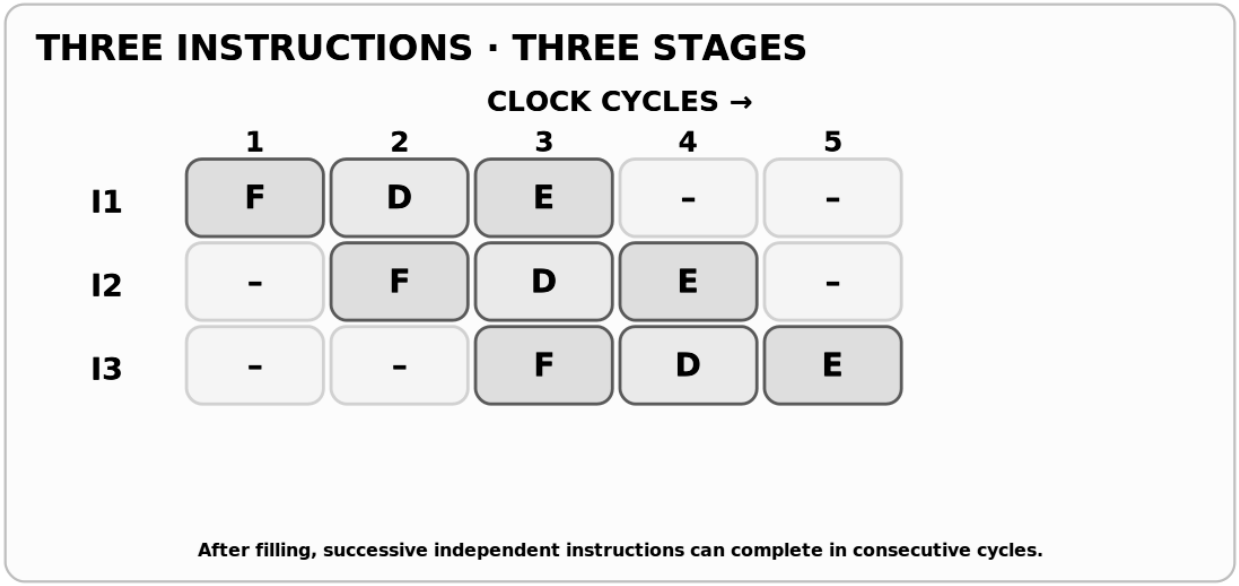
Branches, data dependencies and waiting for memory may cause stalls or pipeline flushing; pipelining does not make every instruction individually three times faster.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

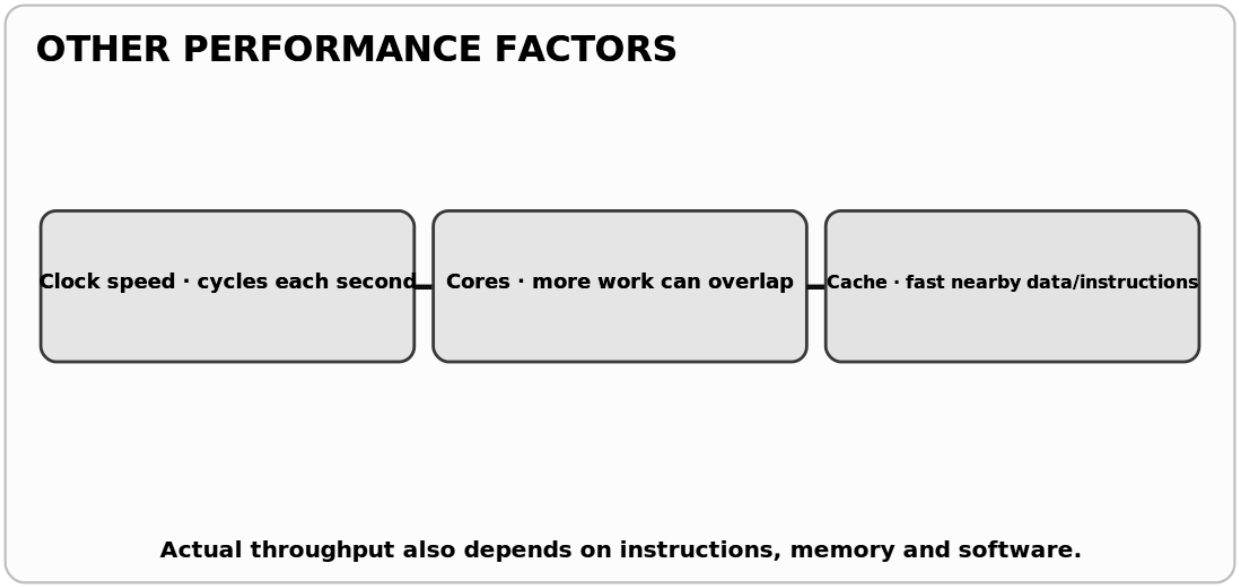
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



THREE INSTRUCTIONS · THREE STAGES: After filling, successive independent instructions can complete in consecutive cycles.

New Learning



OTHER PERFORMANCE FACTORS: Actual throughput also depends on instructions, memory and software.

I DO — WORKED EXAMPLE

Follow the worked method

Trace two independent instructions through a 3-stage F–D–E pipeline.

Predict first

Cycle 1 fetch I1. In cycle 2 decode I1 while fetching I2.

Explain and check

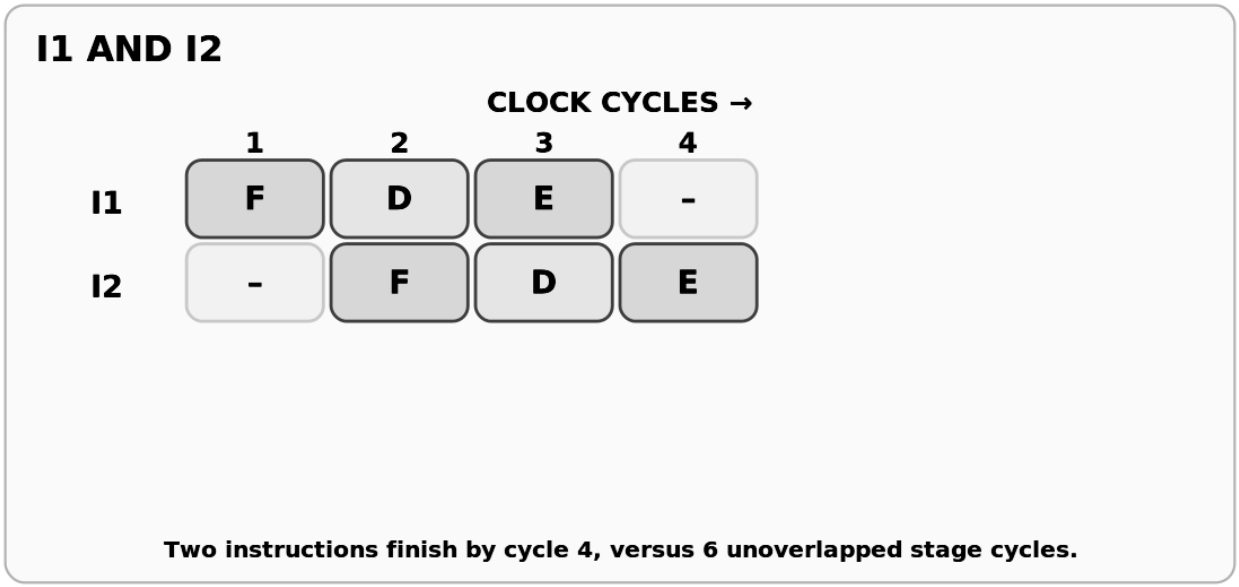
Cycle 3 executes I1 and decodes I2; I2 executes in cycle 4.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace two independent instructions through a 3-stage F–D–E pipeline.

Cycle 1 fetch I1. In cycle 2 decode I1 while fetching I2.



I1 AND I2: Two instructions finish by cycle 4, versus 6 unoverlapped stage cycles.

WE DO — GUIDED PRACTICE

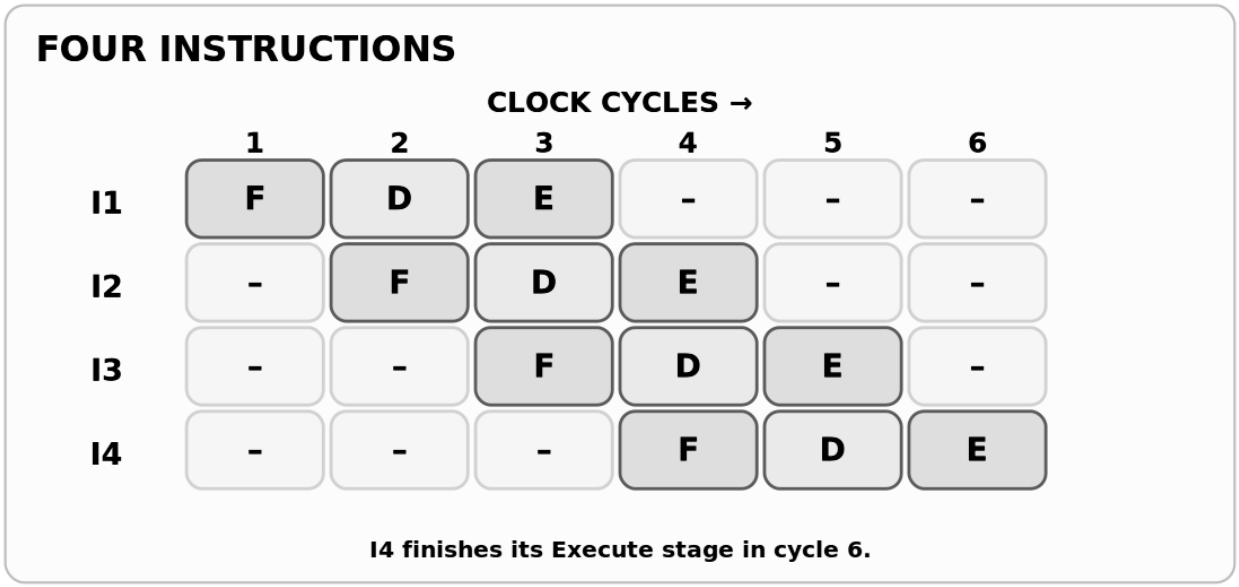
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace four independent instructions through the same pipeline. In which cycle does I4 finish?

Begin a new fetch on each clock cycle, while older instructions advance to Decode or Execute.



FOUR INSTRUCTIONS: I4 finishes its Execute stage in cycle 6.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Begin a new fetch on each clock cycle, while older instructions advance to Decode or Execute.

DISCUSS — Compare approaches, walk through one test input, and explain how you would detect or correct a mistake.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State two processor performance factors and explain one way each might improve throughput.

Question 2

For a three-stage F–D–E pipeline with three independent instructions, write the stages occupied in cycles 1–5 and the completion cycle for I3.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why doubling clock speed does not necessarily double an application's performance. Include one pipeline hazard.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Pipelining always reduces the number of stages required for an individual instruction.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain and check

Instruction four starts Fetch in cycle 4, Decodes in 5 and Executes in 6.

IMPROVE: Explain one correction to your own reasoning.